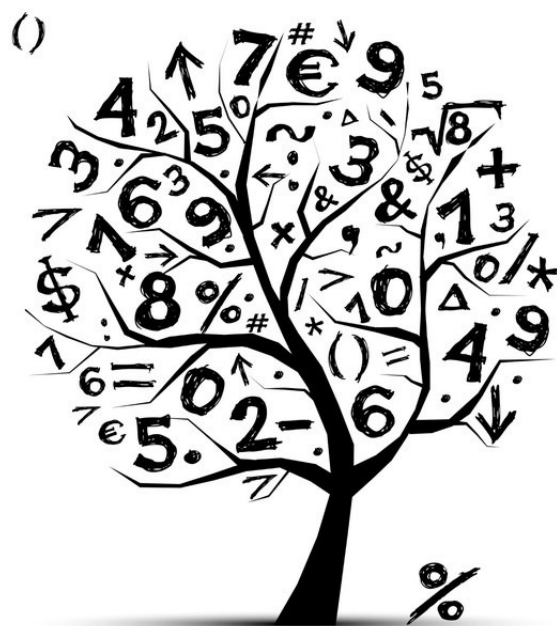


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Inversion of matrices



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1 Computation of the inverse

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Gauss–Jordan method (idea). To compute A^{-1} , we build the augmented matrix

$$(A \mid I)$$

and we apply **elementary row operations** to transform the left block into the identity matrix. If we obtain

$$(I \mid B),$$

then $B = A^{-1}$.

Esempio 1: inverse matrix via Gauss–Jordan elimination

Let us consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We compute the inverse matrix A^{-1} with the method of Gauss–Jordan.

$$\begin{pmatrix} 2 & 1 & \mid & 1 & 0 \\ 1 & 1 & \mid & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & \mid & 0 & 1 \\ 2 & 1 & \mid & 1 & 0 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\begin{pmatrix} 1 & 1 & \mid & 0 & 1 \\ 0 & -1 & \mid & 1 & -2 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 1 & \mid & 0 & 1 \\ 0 & 1 & \mid & -1 & 2 \end{pmatrix} \quad (R_2 \leftarrow -R_2)$$

$$\begin{pmatrix} 1 & 0 & \mid & 1 & -1 \\ 0 & 1 & \mid & -1 & 2 \end{pmatrix} \quad (R_1 \leftarrow R_1 - R_2)$$

Since the left block is the identity matrix, we obtain:

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Esempio 2: inverse matrix via Gauss–Jordan elimination

Let us consider

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

We compute the inverse matrix A^{-1} with the method of Gauss–Jordan.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -4 & 1 & -2 & 0 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - 2R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -4 & 1 & -2 & 0 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 - 2R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 5 & -2 & 4 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 + 4R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{array} \right) \quad (R_3 \leftarrow \frac{1}{5}R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{array} \right) \quad (R_1 \leftarrow R_1 + 2R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{array} \right) \quad (R_2 \leftarrow R_2 - R_3)$$

Since the left block is the identity matrix, we obtain:

$$A^{-1} = \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix}.$$